1 Sequences and Series

Central question. How do repeated steps produce patterns, and how can we predict both a later value and a cumulative total?

1.1 A sequence is an ordered list

A sequence assigns a number a_n to each positive integer n. For example, $a_n = 3 + 2(n-1)$ produces $3, 5, 7, 9, \ldots$. We distinguish the *term* a_n from a *partial sum* $S_n = a_1 + \cdots + a_n$.

1.2 Arithmetic sequences: repeated addition

An arithmetic sequence has a fixed common difference d:

$$a_{n+1} = a_n + d, \qquad \boxed{a_n = a_1 + (n-1)d}.$$

Example 1.1. A theater's first row has 12 seats, with two more seats in each successive row. Row 15 has $12 + 14(2) = 40$ seats.

Adding the first and last terms in reverse order gives

$$\boxed{S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}(2a_1 + (n-1)d)}.$$

1.3 Geometric sequences: repeated multiplication

A geometric sequence has a fixed common ratio q:

$$a_{n+1} = qa_n, \qquad \boxed{a_n = a_1 q^{n-1}}.$$

When $q > 1$, positive terms grow exponentially. When $0 < q < 1$, they decay toward zero.

Example 1.2. A population of 500 organisms increases by 8 percent each generation. The model is $a_n = 500(1.08)^{n-1}$ if generation 1 is the initial observation.

For $q \neq 1$, multiply the finite sum by q and subtract:

$$S_n = a_1 + a_1 q + \cdots + a_1 q^{n-1}, \qquad \boxed{S_n = a_1 \frac{1 - q^n}{1 - q}}.$$

If $q = 1$, the sum is na_1. The formulas refer to *finite* sums and work for ratios outside the interval $(0, 1)$ as well.

1.4 Sigma notation

The compact expression $\sum_{k=1}^{n} a_k$ means add the terms for integer indices $k = 1, \ldots, n$. For instance,

$$\sum_{k=1}^{4} (2k + 1) = 3 + 5 + 7 + 9 = 24.$$

The symbol $\sum$ tells us to add, while its bounds tell us which indices are included.

1.5 Infinite geometric series

If $|q| < 1$, then $q^n \to 0$. The finite geometric sum approaches

$$\boxed{\sum_{n=0}^{\infty} aq^n = \frac{a}{1-q}}, \qquad |q| < 1.$$

The condition matters: the series $1 + 2 + 4 + \cdots$ does not approach a finite sum.

Example 1.3. A ball bounces to 60% of its previous height. Released from height 5 m, the total vertical travel after infinitely many idealized bounces is $5+2(5)(0.6+0.6^2+\cdots) = 5+10(0.6/0.4) = 20$ m.

1.6 Recursive rules beyond arithmetic and geometric patterns

Not every sequence is arithmetic or geometric. Consider

$$a_{n+1} = \frac{a_n + 6}{2}, \qquad a_1 = 0.$$

The first terms are $0, 3, 4.5, 5.25, \ldots$. Let $e_n = a_n - 6$ denote the difference from the fixed point. Then $e_{n+1} = e_n/2$, so $e_n = -6(1/2)^{n-1}$ and

$$a_n = 6 - 6(1/2)^{n-1}.$$

The original recursion is not geometric, but its *error from equilibrium* is. This is a useful bridge to feedback and dynamical models.

1.7 Exercises

1. Write the first five terms of $a_n = 7 - 3(n-1)$ and find a_{20}.
2. An arithmetic sequence has $a_1 = 4$, $d = 5$. Calculate S_{30}.
3. A geometric sequence has $a_1 = 81$, $q = 1/3$. Find a_6.
4. Compute $\sum_{k=0}^{5} 2(3/2)^k$ exactly.
5. Explain the difference between $\sum_{k=1}^{n} k$ and its final term n.
6. Determine whether $5 + 5/2 + 5/4 + \cdots$ converges and find its sum.
7. A machine loses 12 percent of its value per year. Express its value after n years if it initially costs \$12,000.
8. For $a_{n+1} = (a_n + 10)/2$, $a_1 = 2$, find the fixed point and an explicit formula for a_n.